

SKANDA BS

Fourth Year Undergraduate – Bangalore Institute of Technology, Bengaluru

bsskanda@gmail.com | +91-8197228626 | GitHub | LinkedIn

Professional Summary

Software developer and AI engineer with strong foundations in programming, OOP, DSA, DBMS, backend development, and full-stack application development. Experienced in building LLM-powered applications, multi-agent workflows, RAG pipelines, and AI-driven software systems using Python, Django, REST APIs, LangChain, and LangGraph. Interested in autonomous software engineering, intelligent retrieval, AI system evaluation, and scalable AI platforms.

Technical Skills

- **Programming:** C, C++, Python, JavaScript, TypeScript
- **AI/ML:** Machine Learning, LLMs, RAG.
- **Agentic AI:** LangGraph, LangChain, Ollama, Multi-Agent Systems, HuggingFace
- **Backend:** Django, Django REST Framework, REST API
- **Frontend:** React.js, HTML, CSS, Tailwind CSS
- **Databases & Retrieval:** MySQL, MongoDB, FAISS, SQLite.
- **Tools:** Git, GitHub, Docker, Render, Vercel.
- **Software Engineering:** OOP, Data Structures & Algorithms, DBMS.

Projects

AgentShield – AI Agent Security & Governance Platform

- Building a security and governance layer for AI agents that intercepts tool and API requests and evaluates them before execution.
- Designing a gateway-based architecture with agent permissions, prompt-injection detection, PII detection, risk scoring, and policy-based ALLOW, BLOCK, or HITL decisions.
- Implementing audit logging and a kill-switch mechanism to monitor agent actions and provide emergency control over high-risk agent activity.
- Developing integrations for LangChain, LangGraph, and MCP to provide a reusable security layer across different agentic AI applications.

Tech Stack Used: Python, FastAPI, LangGraph, LangChain, MCP, PostgreSQL, Docker, LLMs, Render, Vercel, Neon

Debate AI – Multi-Agent Answer Reliability System

- Built a multi-agent answer reliability system that routes user queries to GPT, Gemini, Grok, and Claude and aggregates their responses for comparative analysis.
- Designed a Django REST Framework backend to process user queries, integrate multiple LLM APIs, and orchestrate model responses through a structured workflow.
- Developed a React-based interface for displaying responses from multiple AI agents and visualizing the debate process through a clear timeline/thread.
- Implemented a multi-model workflow to collect, organize, and compare responses from multiple LLM agents within a unified interface.

Tech Stack Used: Python, Django REST Framework, React, GPT, Gemini, Grok, Claude, Ollama.

Live Demo: debate-ai-1-frontend.onrender.com

IndicRAGHal: Cross-Lingual Hallucination Detection Framework for RAG Systems in Indian Languages

Ongoing

- Building the first benchmark and detection framework for measuring hallucinations in RAG systems across Indian languages (Hindi, Kannada) – a gap unaddressed by existing English-only tools like RAGTruth and LettuceDetect.
- Designing a cross-lingual retrieval pipeline using FAISS and LaBSE sentence embeddings over the BPCC multilingual corpus, paired with BharatGen Param2 and Gemini for generation.
- Fine-tuning mDeBERTa-v3 as a 3-way NLI classifier (Entailment / Contradiction / Neutral) to detect unsupported claims at the sentence level in Indic-language outputs.
- Leading construction of a hand-annotated multilingual dataset using 2-annotators, majority-consensus labeling protocol to ensure label quality across 2 languages.

Tech Stack: Python, PyTorch, Hugging Face Transformers, mDeBERTa-v3, FAISS, Sentence-Transformers (LaBSE), Gemini API, Scikit-learn, Pandas, NLTK

Education

Bangalore Institute of Technology

Sept 2023 – Present

B.E. in Robotics and Artificial Intelligence | CGPA: 8.83/10

Coursework: Machine Learning, Artificial Intelligence, Algorithms, ROS

PES PU College

2021 – 2023

Pre-University (Science) | 95.3%

Achievements

- Solved 200+ Data Structures and Algorithms problems, demonstrating strong problem-solving and coding proficiency.
- Solved 120+ Database Management Systems problems, demonstrating strong understanding of database concepts and problem-solving.
- **4th Place – Casethon, BNM Institute of Technology:** Placed top-4 in a technical case-study competition, delivering a rapid AI-based solution design under time pressure.
- **Top 5 – Anveshna Idea Pitching Competition:** Pitched an AI-driven product concept to a panel of industry judges, earning recognition for innovation and feasibility.